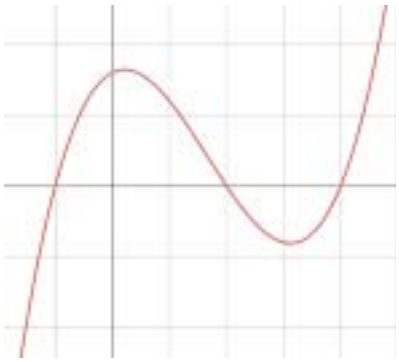
Question	Answer	Marks	Partial Marks
8(a)(ii)	Correct labelled sketch positive cubic Crossing x-axis at –1, 2 and 4 only Crossing y – axis at 8 only 	**4**	**B1** for positive cubic **B2** for three intercepts only with x -axis labelled at – 1, 2 and 4 or **B1** for 1 or 2 correctly labelled x – intercepts **B1** for a single intercept on y-axis labelled at 8 but not if line $y = 8$
8(b)	$3x^2 - 10x - 8$ [= 0]	**M3**	**B2** for derivative = $3x^2 - 10x + 2$ isw OR **B1** for derivative with $3x^2$ or $-10x$ given in expression isw **M1dep on B1** for *their* first derivative = 10
	$x = 4$ and $x = -\frac{2}{3}$	**B1**	
	(4, 0) and $\left(-\frac{2}{3}, \frac{112}{27}\right)$ oe	**B1**	
	[y =] $10x - 40$ and [y =] $10x + \frac{292}{27}$	**B2**	**B1** for each or for two different equations of the form [y =] $10x + c$ (c must be numeric) or for $c = -40$ and $\frac{292}{27}$
9(a)(i)	$27x^6y^{12}$ final answer	**2**	**B1** for two terms correct in answer e.g. $27x^6y^k$ or $27x^ky^{12}$ or kx^6y^{12} or for correct answer seen then spoilt

Question	Answer	Marks	Partial Marks
9(a)(ii)	$\frac{x^{24}y^{12}}{64}$ final answer	**3**	**B2** for final answer with two correct elements or final answer $\frac{64}{x^{24}y^{12}}$ or $\frac{64^{-1}}{x^{-24}y^{-12}}$ or better or for correct answer seen or **B1** for 64 or x^{24} or y^{12} seen in final answer or final answer $\frac{k}{x^{-24}y^{-12}}$ or **M1** for first correct step seen eg $\left(\frac{x^{16}y^{8}}{16}\right)^{\left[\frac{3}{2}\right]}$ or $\left(\frac{4}{x^{8}y^{4}}\right)^{[-3]}$ or $\left(\frac{4096}{x^{48}y^{24}}\right)^{\left[-\frac{1}{2}\right]}$
9(b)(i)	$(x + 3)(x – 3)$ final answer	**1**	
9(b)(ii)	$\frac{x+3}{2y+5}$ final answer	**3**	**M2** for $(x – 3)(2y + 5)$ or **M1** for $2y(x – 3) + 5(x – 3)$ or $x(2y + 5) – 3(2y + 5)$
9(c)	$5x^2 + 4x – 20$ [= 0] oe or $5y^2 – 78y + 221$ [= 0] oe	**M2**	**M1** for $7 – 2x = 5x^2 + 2x – 13$ oe seen or $y = 5\left(\frac{7-y}{2}\right)^2 + 2\left(\frac{7-y}{2}\right) - 13$ oe seen
	$\frac{-4 \pm \sqrt{(4)^2 - 4(5)(-20)}}{2(5)}$ oe or $-\frac{4}{10} \pm \sqrt{4 + \left(\frac{4}{10}\right)^2}$ oe	**M2**	**FT** *their* 3-term quadratic or **M1** for $\sqrt{(4)^2 - 4(5)(-20)}$ or better or for $\frac{-4+\sqrt{q}}{2\times 5}$ or $\frac{-4-\sqrt{q}}{2\times 5}$ or for $\left(x + \frac{4}{10}\right)^2$ oe
	$x = 1.64$ $y = 3.72$ and $x = – 2.44$ $y = 11.88$	**B2**	**B1** for one correct pair or both x-values correct or both y – values correct